Luke Belliveau

Platform Engineer

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Platform engineer with a decade of experience building systems at enterprise scale. Builds the platforms product teams build on: delivery platforms, cloud infrastructure, data platforms and the developer tooling around them.

Experience

Navalia

2023 – Present

Consultant Software Engineer

Agentic Infrastructure Platform

2025 – Present

- Building a developer "platform in a box": a single installation sets up cloud infrastructure, Kubernetes clusters, CI/CD pipelines, GitOps, logging and alerting inside the organization's own cloud accounts.
- Teams describe each application in one manifest file. The platform provisions its databases, caches, message queues and secrets, and generates a pipeline that tests, security scans and deploys every push through dev, staging and production, with approval gates and runtime verification.
- Built an AI engineering agent into the platform that investigates alerts across code, builds, logs and live systems, then prepares, tests and opens fixes as pull requests for review.

TypeScript, Python, Terraform, Kubernetes, Argo CD, GCP, AWS, Buildkite, Harness, Claude

Gamification Platform

2025 – 2026

- Designed, built and launched the dual-region GCP/GKE platform behind a large enterprise's gamification product.
- App teams declare what they need in a single YAML file. Across 7 production services, every push to main runs tests, security scans, image builds, provisioning, deployment and end-to-end tests.
- Migrated the platform onto the enterprise's shared multi-tenant clusters, Istio mesh, policy gates and security scanners with zero downtime and no changes required from app teams.
- Launch week (June 2026): 5.8M requests at 99.986% success, peaking at 9,850 requests per minute, across 97,016 game sessions from 89,792 unique customers.

Terraform, GCP, GKE, Istio, Harness, Argo CD, Helm, Cloud SQL, Python, FastAPI

Path Robotics · Robotics Data Platform

2024 – 2025

- Led architecture of a domain-driven data platform for autonomous welding robots that generate laser-scanner point clouds, 3D models and sensor data.
- Replaced tightly coupled pipelines with independently owned data domains, consumer-specific API contracts, and route-level access approved by each domain's owner.
- Built an edge proxy with write buffering and read caching, so factory floors keep operating offline and sync to the cloud as bandwidth allows.
- Moved AI training and robot operations from overnight batch jobs to real-time data, cut mean time to discover and resolve incidents by 80%, and designed for a 100x increase in data volume.

Python, FastAPI, PostgreSQL, RabbitMQ, Terraform, AWS, GitHub Actions

Wendy's · Web Ordering Platform

2023 – 2024

- Full stack engineer on the web ordering platform for Wendy's 7,000+ restaurants.
- Liaison between web and platform engineering during the AWS to GCP migration: dual-repository sync, single-image multi-environment deployments with runtime configuration, and GitOps with Argo CD.
- Contributed to FreshAI, Wendy's AI drive-thru ordering system, which processes tens of thousands of orders daily.
- Kickstarted a Next.js 14 rewrite of the 318K-line ordering application, introducing Server Components, Zustand and domain-driven architecture.

React, Next.js, TypeScript, Redux, Zustand, Docker, Kubernetes, Argo CD, GitLab CI, GCP

NuWave Technologies 2021 – 2023

Consultant Software Engineer

- Led development of the Prizm API Gateway management console on a team of three, from inception to launch.
- Migrated the LightWave Client and Server products from AngularJS to React, reducing source code by 35%.
- Built automated testing pipelines and ran training that raised front-end code coverage from 0% to over 95%.

ZUUT 2020 – Present

Technical Partner

- Built a browser-based CAD tool for planning cannabis grow operations from inception to launch, including a canvas interaction and collision engine for 100+ items (Redux, Firebase).
- Added account management, Amazon affiliate equipment comparison and Contentful publishing.

Grubhub 2018 – 2020

Senior Cloud Infrastructure Engineer

- Owned the notifications microservices (SMS, push, email, fax) for 500K+ daily active users (Java, Spring).
- Extended the core Java service framework for communication in a new Envoy service mesh.
- Designed the architecture for a new notifications platform, approved by the org-wide architecture review council.
- Built a React and Python tool for SREs to view system status and make changes during outages.

ThoughtWorks 2016 – 2018

Senior Software Engineer, Associate Software Engineer

- Delivered client software in health insurance, e-commerce and cybersecurity, including a React and Java insurance portal, a Vue analytics dashboard, and data pipelines for Kroger's e-commerce platform.
- Built a Go backend-for-frontend that aggregated and transformed logs before serving them to the client.
- Ran front-end training and a global community of React and Vue developers, and mentored new hires.

Skills

Languages	Python, TypeScript, JavaScript, Kotlin, Java, Go, SQL
Infrastructure	Terraform, Kubernetes, Helm, Istio, Docker, GCP, AWS
Delivery	Buildkite, Argo CD, GitHub Actions, GitLab CI, Harness, Jenkins, Spinnaker
Data	PostgreSQL, RabbitMQ, Cassandra, MongoDB, Firebase
Backend & web	FastAPI, Spring Boot, React, Next.js, Redux, Zustand
Testing & tracing	pytest, Vitest, Jest, JUnit, Playwright, Cypress, OpenTelemetry

Projects

Selenium.dev 2019 · New website for the browser automation project, built with Jekyll and Hugo.

Devil's Dice 2018 · Mobile party game; 10,000+ downloads and a 4.5-star rating on Google Play.

SATURATION 2017 · Official BROCKHAMPTON app for early releases; 10,000+ downloads and a 5-star rating.

Education

B.S. Computer Science, University of Connecticut 2011 – 2016